

System Workflow Documentation

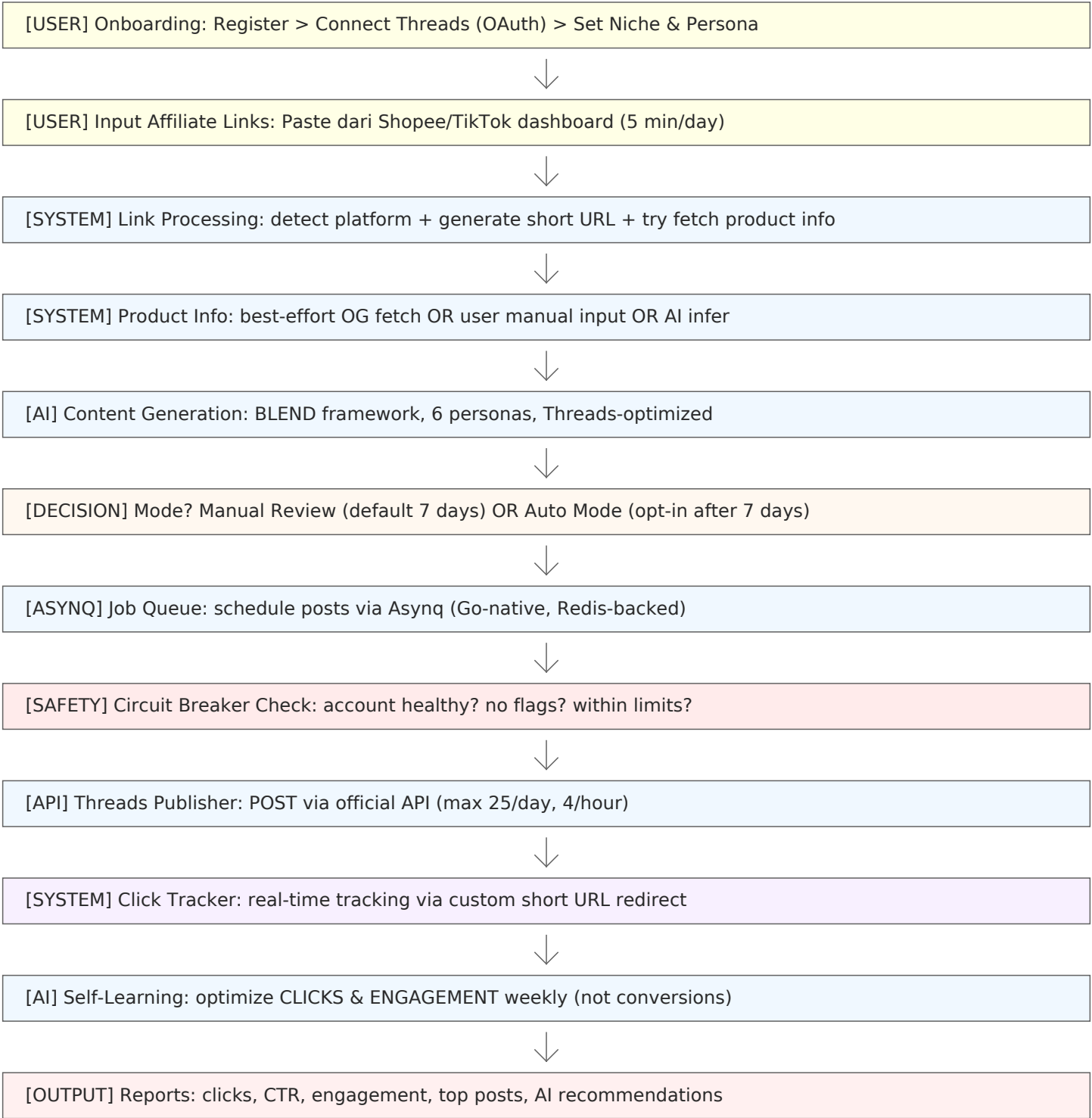
Threads Affiliate Automation Platform Shopee + TikTok Shop | Manual Link Input

v2.0 - Post-Audit (All Critical Issues Fixed)

Versi: 2.0.0 (Audit-Fixed)
Tanggal: 06 May 2026
Referensi: PRD v2.0.0
Job Queue: Asynq (Go-native)
Optimization Target: Clicks & Engagement

1. High-Level System Pipeline (Updated)

End-to-end flow dengan semua fixes applied:

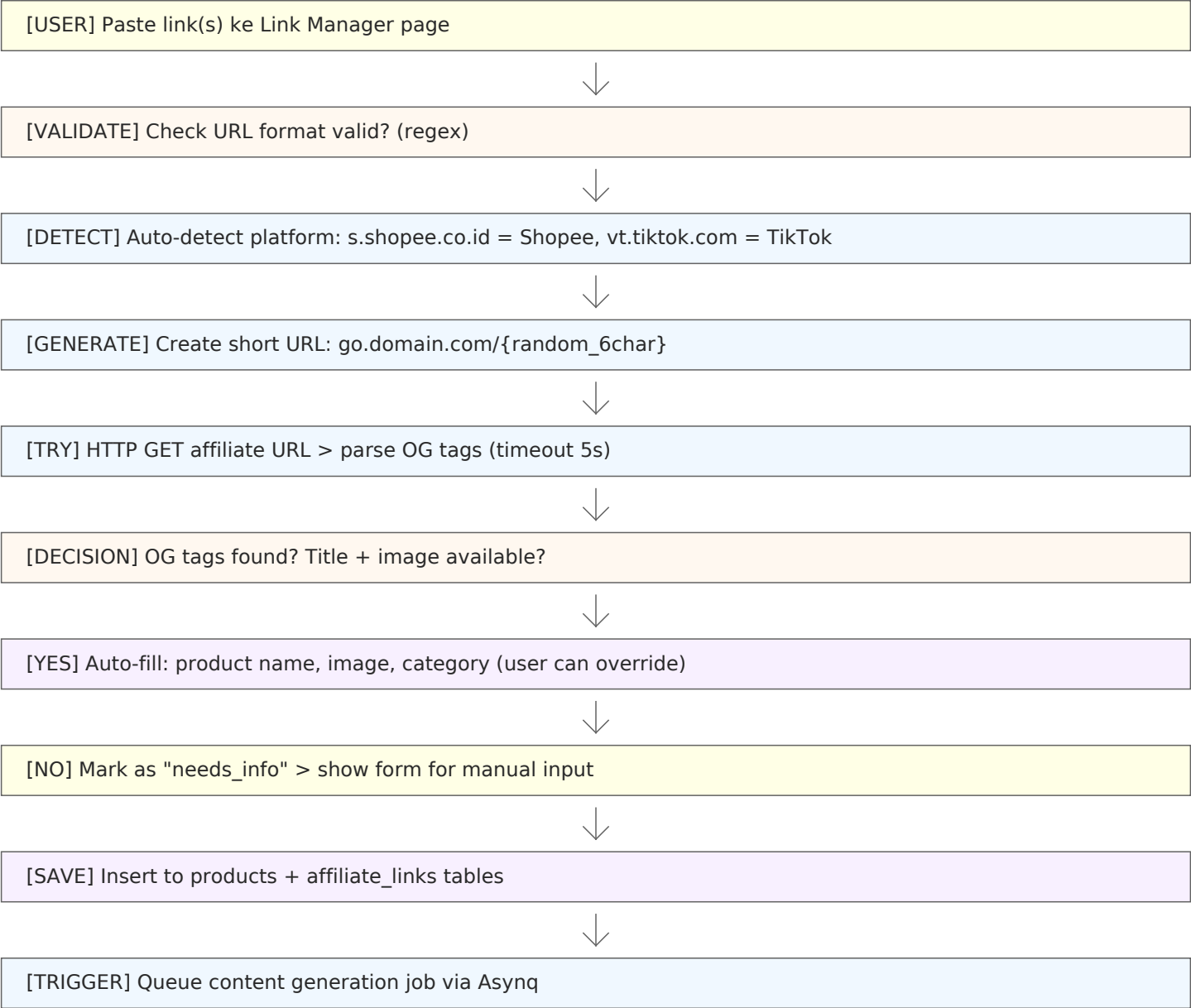


Legend:
Kuning = User action (manual, ~5 min/day)
Biru = System automation
Merah muda = Safety check (circuit breaker)
Ungu = Data storage/tracking

2. Workflow: Link Input & Product Info (Fixed)

FIXED: Product info = best-effort. User manual input = primary. Auto-fetch = bonus.

2.1 Link Input Flow



2.2 Platform Detection Rules

URL Pattern	Platform	Confidence
s.shopee.co.id/*	Shopee	100%
shopee.co.id/*?af_id=*	Shopee	100%
vt.tiktok.com/*	TikTok Shop	100%
tiktok.com/*/product/*	TikTok Shop	95%

tokopedia.link/*	Tokopedia (future)	100%
Other	Ask user	0%

2.3 Product Info Fallback Chain

1.

Try: HTTP GET > follow redirects > parse og:title, og:image (5s timeout)
2.

If fail: AI infer category from URL path/slug (e.g., "/skincare/" = skincare)
3.

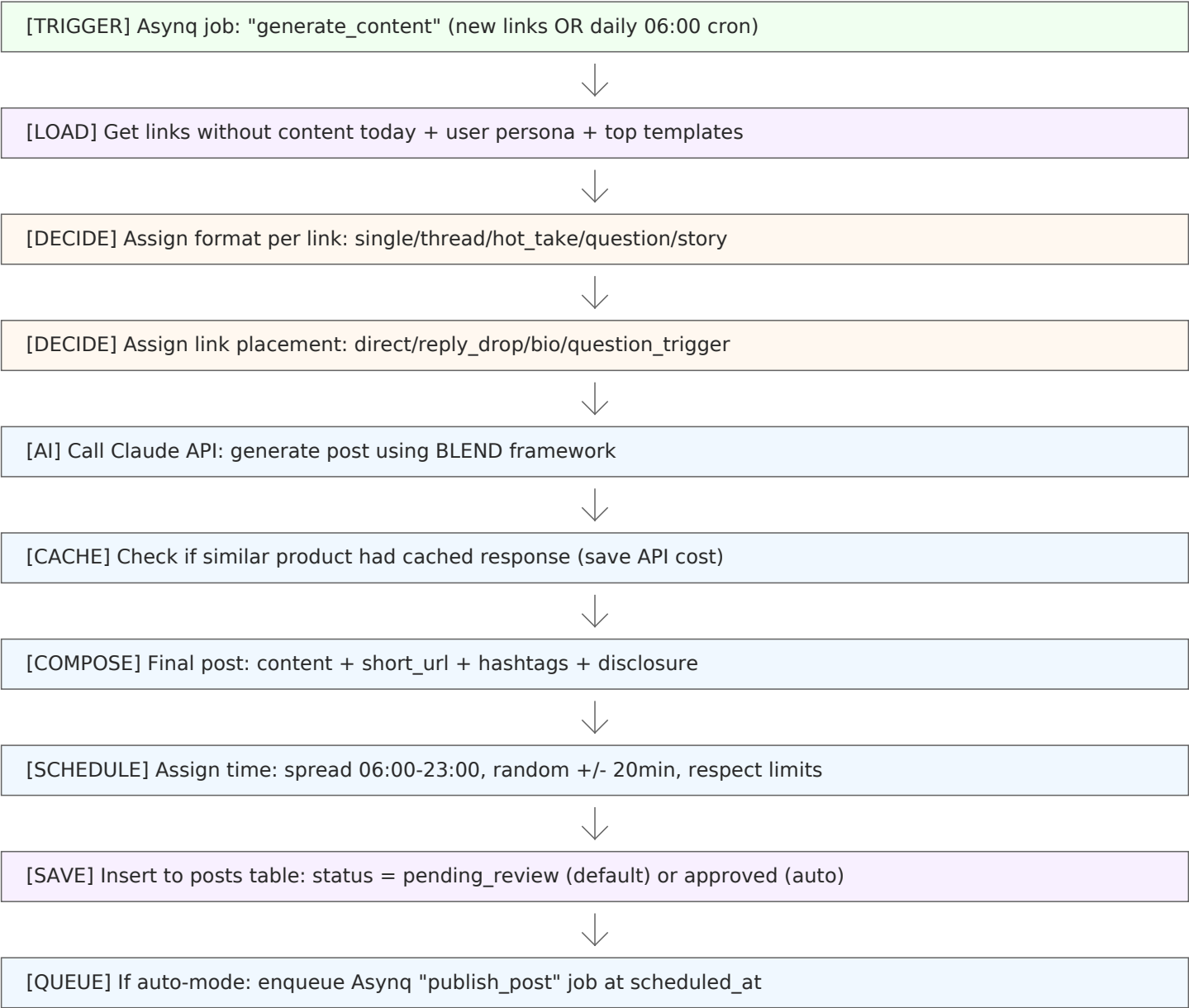
If fail: mark "needs_info", prompt user for manual input
4.

If user skips: AI generates generic content (still works, less specific)
- Success rate estimate: Shopee ~60%, TikTok ~40%. User manual input always available.

3. Workflow: AI Content Generation (Fixed)

Uses Asynq job queue (Go-native). Triggered by new links or daily cron.

3.1 Flow Diagram



3.2 Content Rules (Consistent with PRD)

Rule	Value	Enforced By
Max posts/day/account	25	Scheduler (hard limit)
Max posts/hour	4	Scheduler (hard limit)
Max affiliate posts in row	2	Content planner
Content mix	60% value / 40% affiliate	Content planner

Min interval	45 minutes	Scheduler
Affiliate disclosure	Include in every affiliate	Post composer
No duplicate angle	Same product+angle block	Dedup check

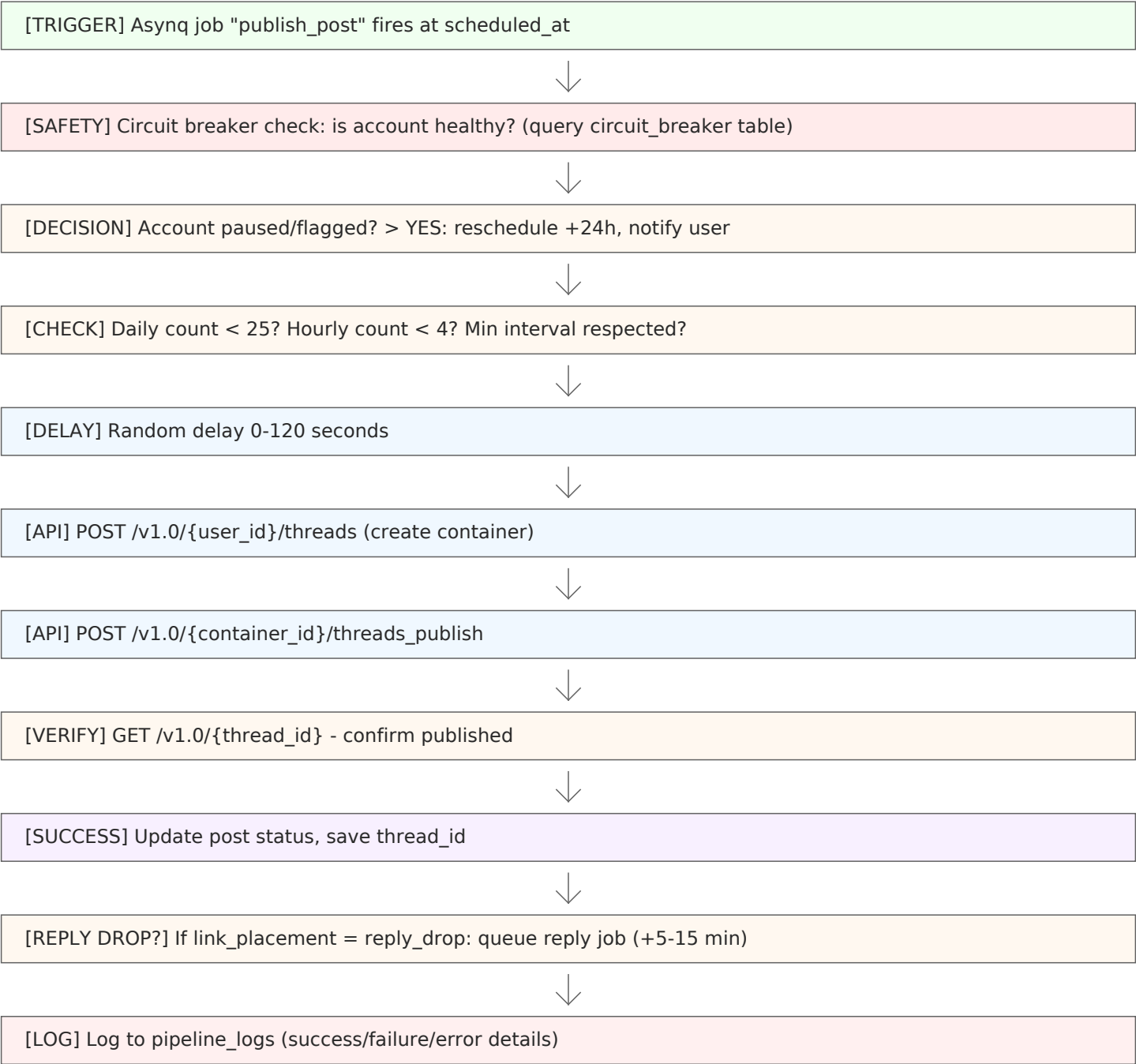
3.3 Asynq Job Types

Job Type	Trigger	Retry Policy	Timeout
generate_content	New links / daily cron	3 retries, 30s backoff	60s
publish_post	Scheduled time reached	2 retries, 60s backoff	30s
reply_drop	5-15 min after publish	2 retries, 30s backoff	30s
check_replies	Every 10 min (periodic)	1 retry	30s
collect_analytics	Every 4 hours (periodic)	2 retries	60s
health_check_links	Every hour (sample 10%)	1 retry	30s
weekly_learning	Monday 02:00	3 retries	300s

4. Workflow: Publishing (with Circuit Breaker)

Publishing flow dengan safety checks terintegrasi.

4.1 Flow Diagram



4.2 Circuit Breaker Logic (NEW)

Event	Trigger	Action	Auto-Recovery
Post rejected (content)	1 rejection	Log + skip post	Immediate (next post OK)
Multiple rejections	3 in 24h	Pause account 24h	Auto-resume after 24h

Account restricted by M	API 403 response	Pause account 48h	Manual review needed
Multiple accounts flagg	2+ accounts in 24	PAUSE ALL accounts	Manual review needed
API app rate limited	HTTP 429 from Me	Emergency stop	Contact Meta support

4.3 Threads API Sequence (Detailed)

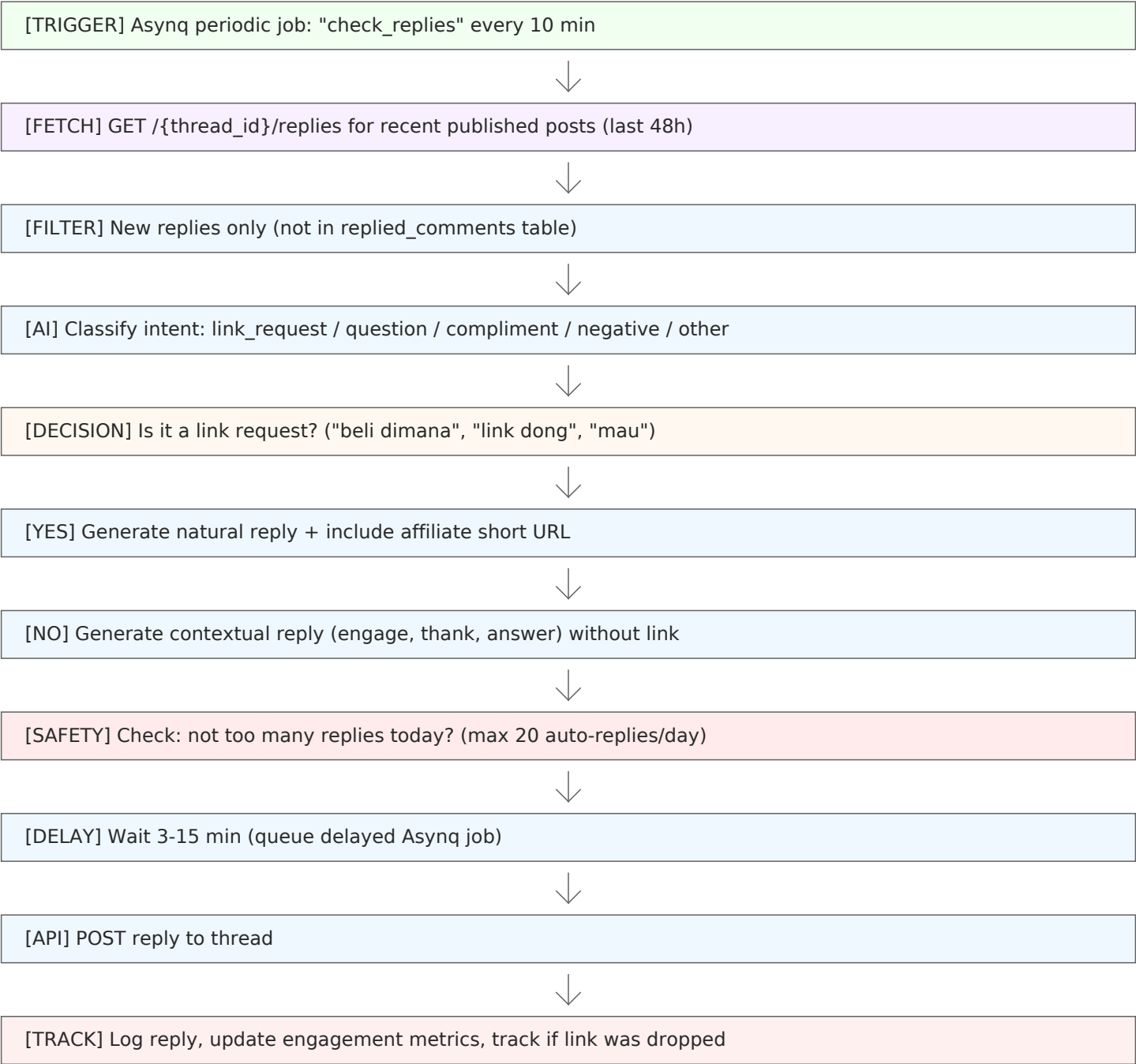
Exact API calls per publish:

- Step 1: POST /v1.0/{user_id}/threads Body: {"media_type":"TEXT","text":"{content}"}
Response: {"id":"container_123"}
- Step 2: POST /v1.0/{user_id}/threads_publish Body: {"creation_id":"container_123"} Response: {"id":"thread_456"}
- Step 3: GET /v1.0/thread_456?fields=id,text,timestamp Verify: status = published

5. Workflow: Engagement & Auto-Reply

Monitor replies, auto-respond with AI. Key for natural link dropping.

5.1 Flow Diagram



5.2 Safety Limits for Auto-Reply

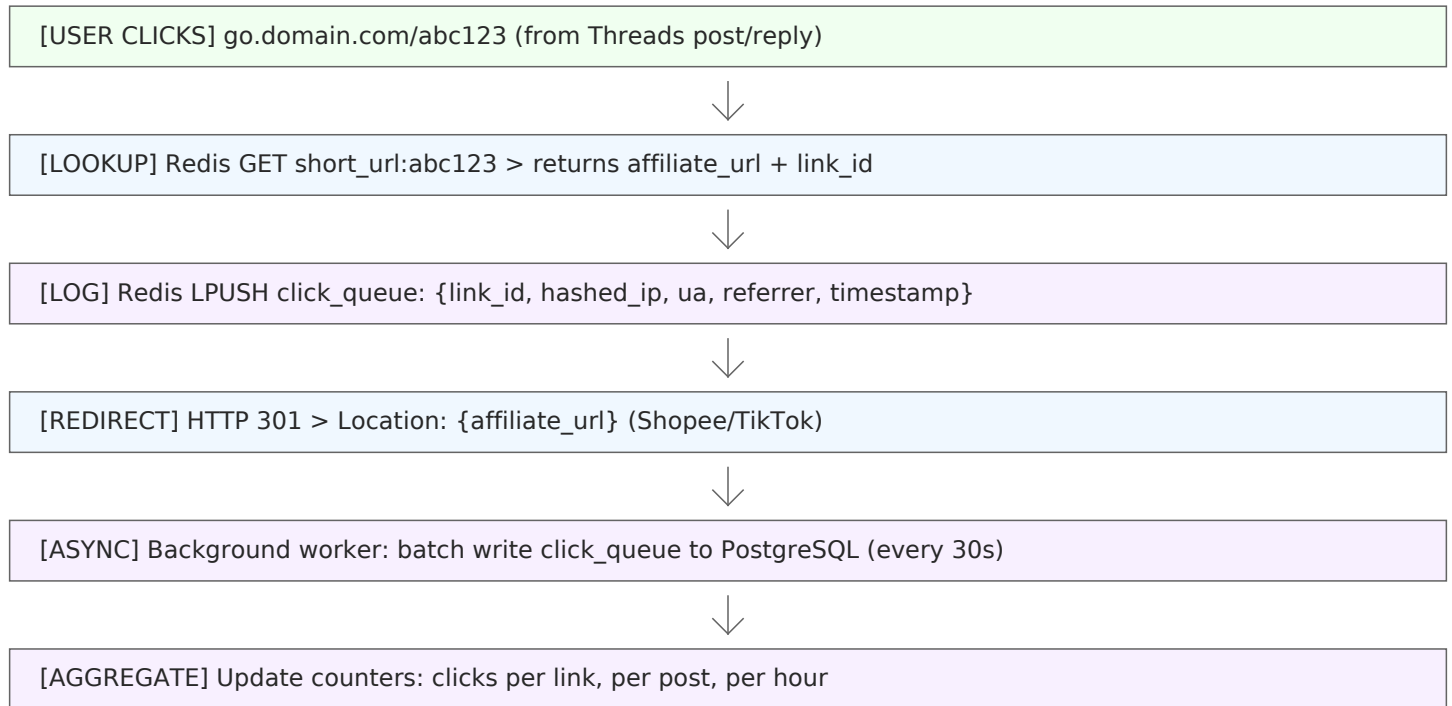
Limit	Value	Rationale
Max auto-replies/day	20	Avoid spam behavior
Max replies with links/day	10	Conservative link dropping

Min delay before reply	3 minutes	Human-like response time
Max delay before reply	15 minutes	Still timely
Reply to same user max	2 per thread	Avoid harassment feel
No reply to negative comm	Skip or generic on	Avoid escalation

6. Workflow: Click Tracking (Real-Time)

Custom URL shortener with sub-50ms redirect latency.

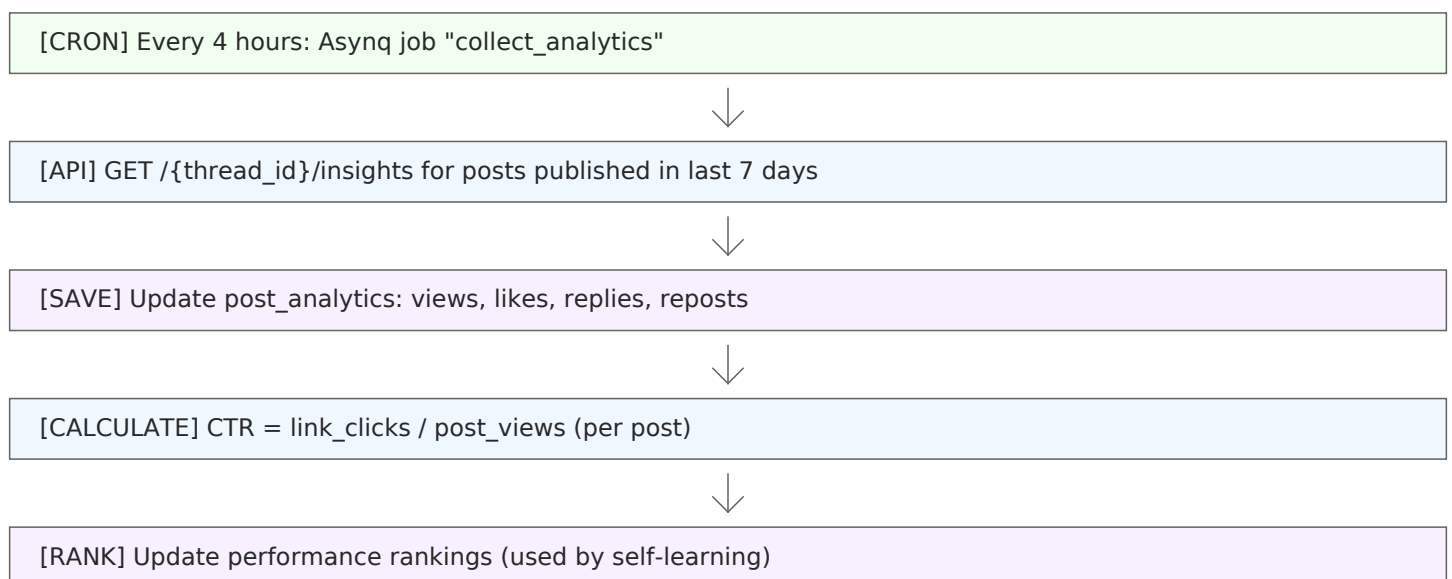
6.1 Click Flow



Performance target: < 50ms p99 redirect latency (Redis lookup + 301 redirect)

Privacy: IP addresses are SHA-256 hashed before storage. Raw IPs never persisted.

6.2 Analytics Collection



7. Workflow: Self-Learning (Fixed)

FIXED: AI optimizes CLICKS & ENGAGEMENT only. NOT conversions.
Conversion data = optional bonus if user provides it.

[TRIGGER] Monday 02:00 WIB: Async job "weekly_learning"



[COLLECT] All posts from last 7 days + their clicks + engagement metrics



[SEGMENT] Group by: persona, format, time, platform, category, hook type



[RANK] Sort by CTR (clicks/views) within each segment



[ANALYZE] AI: compare top 20% vs bottom 20% - extract patterns



[UPDATE] Adjust content_templates scores + generation weights



[PLAN] Create 1 A/B test for next week (single variable)



[REPORT] Generate weekly insight > push to user dashboard

8. System Schedule (Consistent with PRD)

8.1 Daily Automated Jobs

Time (WIB)	Asynq Job	Type	Duration
02:00	db_backup	System	~5 min
06:00	generate_content (daily batch)	AI	~10 min
06:00-23:00	publish_post (per scheduled post)	API	~30s each
*/10 min	check_replies	AI	~2 min
*/4 hours	collect_analytics	API	~5 min
*/1 hour	health_check_links (sample 10%)	System	~1 min
23:00	daily_report	System	~3 min

8.2 Weekly Jobs

Day/Time	Asynq Job	Purpose
Senin 02:00	weekly_learning	Analyze performance, update AI weights
Senin 03:00	ab_test_plan	Design experiment for this week
Senin 08:00	weekly_report	Send comprehensive report to user
Jumat 02:00	template_refresh	Generate new templates from winners
Minggu 02:00	account_health_check	Verify all accounts active

8.3 User-Triggered Jobs

User Action	Asynq Job Queued	Priority	Timing
Paste new links	generate_content	High	Within 2 min
Approve post	publish_post (at scheduled_at)	Normal	At schedule
Force publish now	publish_post (immediate)	High	Within 1 min
Edit post	Update in queue	Normal	Immediate
Change persona	regenerate_content	Low	Within 10 min

8.4 Posting Distribution (25 posts/day)

Window	Posts	Type Mix	Interval
06:00-09:00	4	2 value + 2 affiliate	~45 min
09:00-12:00	5	3 value + 2 affiliate	~36 min

12:00-15:00	5	2 value + 3 affiliate (peak)	~36 min
15:00-18:00	4	3 value + 1 affiliate	~45 min
18:00-21:00	5	2 value + 3 affiliate (peak)	~36 min
21:00-23:00	2	2 value (wind down)	~60 min

Total: 25 posts = 15 value (60%) + 10 affiliate (40%). All times +/- 20 min random.

9. Error Handling & Recovery (Updated)

9.1 Link Input Errors

Error	Detection	User Experience
Invalid URL	Regex validation	Red highlight + "Invalid URL format"
Duplicate link	DB unique constraint	"Link already exists" + show existing
Unknown platform	No pattern match	Dropdown: "Select platform manually"
OG fetch failed	HTTP timeout/error	Show manual input form (not blocking)
Link expired/404	Health check	Yellow badge + "Link may be expired"

9.2 Publishing Errors

Error	Asynq Handling	Circuit Breaker
401 Token expired	Retry after token refresh	No trigger
429 Rate limited	Retry in next hour window	No trigger (expected)
400 Content rejected	No retry, mark failed	Count toward 3/24h limit
500 Server error	Retry 2x with 60s backoff	No trigger
Network timeout	Retry 2x with 30s backoff	No trigger
403 Account restricted	No retry, pause account	TRIGGER: pause 48h

9.3 AI Service Errors

Error	Handling	Fallback
Claude API timeout	Retry 3x, 30s backoff	Use cached template
Claude API 429	Queue for next minute	Delay generation batch
Invalid AI response	Retry with different prompt	Use generic template
AI generates hard-sell	Content filter catches it	Regenerate with stricter prompt

9.4 Data Recovery

- PostgreSQL: daily backup to S3, 30-day retention, point-in-time recovery
- Redis: AOF persistence, no click data loss on restart
- Asynq jobs: persisted in Redis, survive service restart
- Published posts: thread_id saved, can verify/reconcile with Threads API
- Click logs: batch-written every 30s, max 30s data loss on crash

10. Deployment & Monitoring

10.1 Docker Services

Service	Image	Port	Replicas
frontend	nginx:alpine + React build	3000	1
api	Custom Go binary	8080	1-2
ai-service	Python FastAPI	8081	1
url-shortener	Custom Go binary	8082	1-2
asynq-worker	Custom Go binary	N/A	1-3
asynq-scheduler	Custom Go binary	N/A	1
postgresql	postgres:16-alpine	5432	1
redis	redis:7-alpine	6379	1
nginx	nginx:alpine (reverse proxy)	80/443	1
prometheus	prom/prometheus	9090	1
grafana	grafana/grafana	3001	1

10.2 Monitoring & Alerts

Metric	Alert Threshold	Action
API response time	> 500ms p95	Investigate, scale if needed
URL redirect latency	> 100ms p99	Check Redis, CDN
Asynq queue depth	> 100 pending jobs	Scale workers
Publishing success rate	< 90%	Check Threads API, circuit breaker
AI generation success	< 95%	Check Claude API, fallback to cache
Disk usage	> 80%	Cleanup old logs, expand storage
Circuit breaker triggered	Any trigger	Immediate notification to admin

10.3 CI/CD Pipeline

1. Git push to main > GitHub Actions triggered
2. Run unit tests + linting (Go + Python)
3. Build Docker images (multi-stage)
4. Push to container registry
5. Deploy to staging > run integration tests
6. Manual approval > deploy to production

7. Health check > notify success/failure

11. Workflow Summary (Final)

#	Workflow	Queue	Trigger	Safety Check
1	Link Input	Sync (API)	User paste	URL validation
2	Product Info	Asynq (async)	After link input	Timeout 5s, fallback
3	Content Gen	Asynq	New links / cron	Content filter
4	Publishing	Asynq (scheduled)	At scheduled_at	Circuit breaker
5	Reply Drop	Asynq (delayed)	5-15 min after pub	Rate limit check
6	Auto-Reply	Asynq (periodic)	Every 10 min	Max 20/day limit
7	Click Track	Redis (real-time)	User click	IP hashing
8	Analytics	Asynq (periodic)	Every 4 hours	API rate limit
9	Self-Learning	Asynq (weekly)	Monday 02:00	Min sample size
10	Health Check	Asynq (periodic)	Every hour	Sample 10% only

All Critical Issues Fixed:

C1 (Threads API Compliance): Added Legal section, human-in-loop default, disclosure

C2 (Account Suspension): Circuit breaker system, conservative limits (25/day)

C3 (OG Tag Extraction): Best-effort only, manual input primary

C4 (Bull/Go Mismatch): Replaced with Asynq (Go-native, Redis-backed)

C5 (Revenue Tracking): AI optimizes clicks/engagement, revenue = optional manual input

IMPLEMENTATION READY.

Next step: Phase 0 (Validation) - test Threads API publishing before coding.

Then Phase 1 (MVP) - 3 weeks to first working prototype.